KAIZEN MATHS

Lesson starter

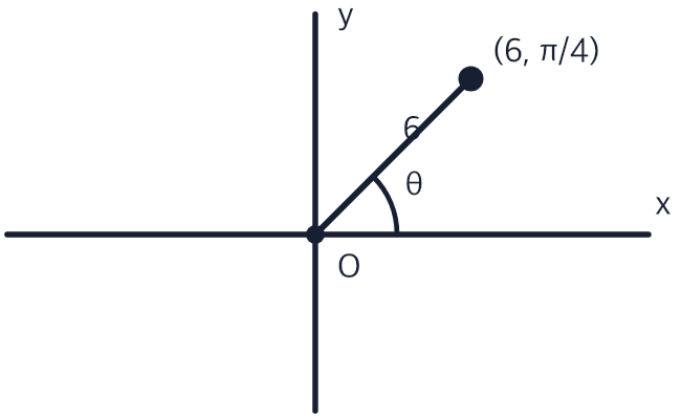
Polar Coordinates and Polar Curves

Polar Coordinates

Name _____

SA1

Convert each polar coordinate into Cartesian form.



Convert the polar coordinate (6, π/4) into Cartesian form.

Show your working

KAIZEN MATHS

Lesson starter

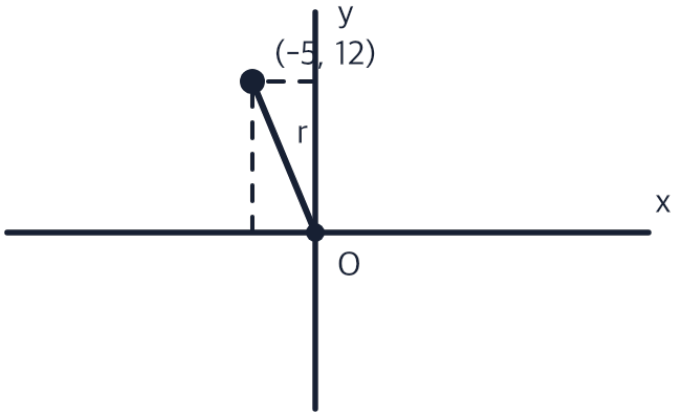
Polar Coordinates and Polar Curves

Polar Coordinates

Name _____

SB1

Convert each Cartesian coordinate into polar form.



Convert (-5, 12) into polar form, using $r > 0$ and $0 \leq \theta < 2\pi$.

Show your working

KAIZEN MATHS

Lesson starter

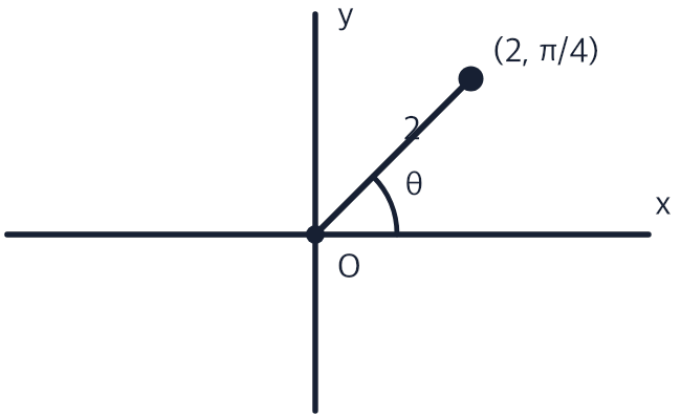
Polar Coordinates and Polar Curves

Polar Coordinates

Name _____

SC1

Give equivalent polar coordinates for each point.



Give two different polar coordinates that represent the same point as (2, π/4).

ALG-294F-L1 · SC

Show your working

KAIZEN MATHS

Lesson starter

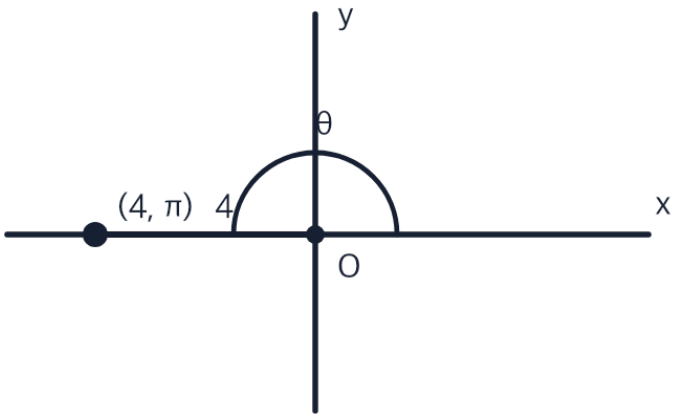
Polar Coordinates and Polar Curves

Polar Coordinates

Name _____

SD1

Convert each polar coordinate into Cartesian form.



Convert the polar coordinate (4, π) into Cartesian form.

ALG-294F-L1 · SD

Show your working

KAIZEN MATHS

Exit ticket

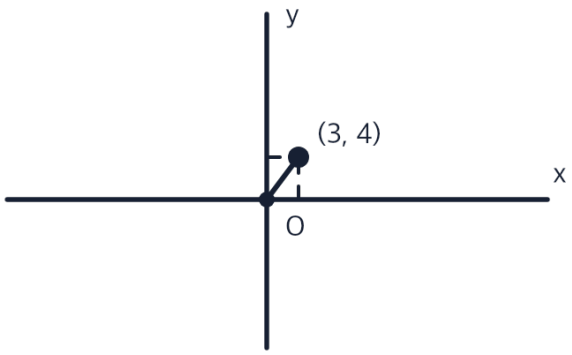
Polar Coordinates and Polar Curves

Polar Coordinates

Name

EA1

Convert each Cartesian coordinate into polar form.



Convert (3, 4) into polar form, using $r > 0$ and $0 \leq \theta < 2\pi$.

Show your working

KAIZEN MATHS

Exit ticket

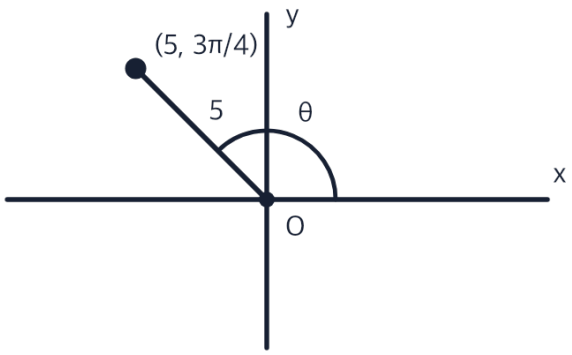
Polar Coordinates and Polar Curves

Polar Coordinates

Name

EB1

Give equivalent polar coordinates for each point.



Give two different polar coordinates that represent the same point as $(5, 3\pi/4)$.

Show your working

KAIZEN MATHS

Exit ticket

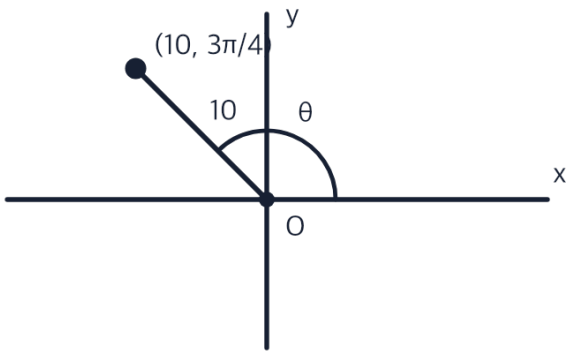
Polar Coordinates and Polar Curves

Polar Coordinates

Name

EC1

Convert each polar coordinate into Cartesian form.



Convert the polar coordinate $(10, 3\pi/4)$ into Cartesian form.

Show your working

KAIZEN MATHS

Exit ticket

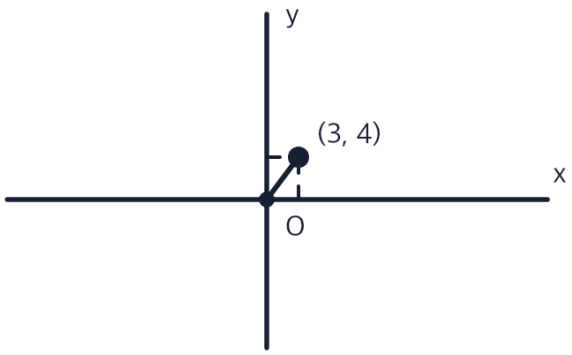
Polar Coordinates and Polar Curves

Polar Coordinates

Name

EA1

Convert each Cartesian coordinate into polar form.



Convert (3, 4) into polar form, using $r > 0$ and $0 \leq \theta < 2\pi$.

Show your working

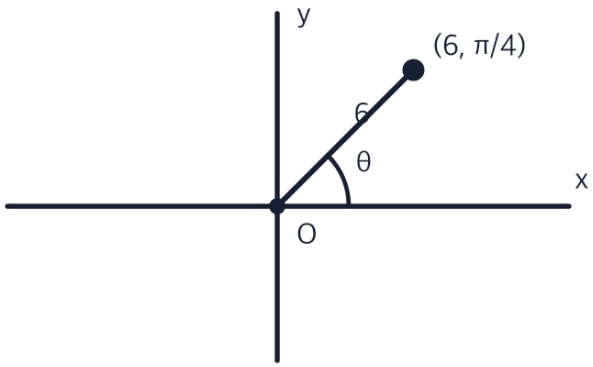
Teacher answers & working

Polar Coordinates and Polar Curves · Polar Coordinates

ALG-294F-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1

Convert each polar coordinate into Cartesian form.



Convert the polar coordinate (6, π/4) into Cartesian form.

Answer:

$(3\sqrt{2}, 3\sqrt{2})$

Working:

Use $x = r\cos\theta$ and $y = r\sin\theta$.

Here r = 6 and θ = π/4.

$x = 6\cos(\pi/4) = 6 \times \frac{\sqrt{2}}{2} = 3\sqrt{2}$.

$y = 6\sin(\pi/4) = 6 \times \frac{\sqrt{2}}{2} = 3\sqrt{2}$.

Therefore the Cartesian coordinate is $(3\sqrt{2}, 3\sqrt{2})$.

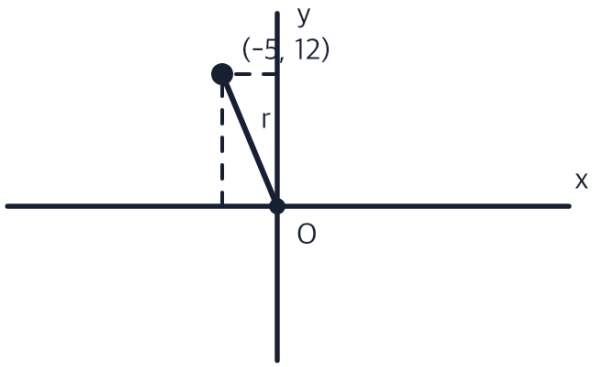
Teacher answers & working

Polar Coordinates and Polar Curves · Polar Coordinates

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SB1

Convert each Cartesian coordinate into polar form.



Convert (-5, 12) into polar form, using $r > 0$ and $0 \leq \theta < 2\pi$.

Answer:

$(13, \pi - \tan^{-1}(12/5))$

Working:

Use $r^2 = x^2 + y^2$.

$r^2 = -5^2 + 12^2 = 169$, so $r = 13$.

Use $\tan\theta = \frac{y}{x} = \frac{12}{-5}$.

Choose the quadrant of the point (-5, 12).

Therefore $\theta = \pi - \tan^{-1}(12/5)$.

The polar coordinate is $(13, \pi - \tan^{-1}(12/5))$.

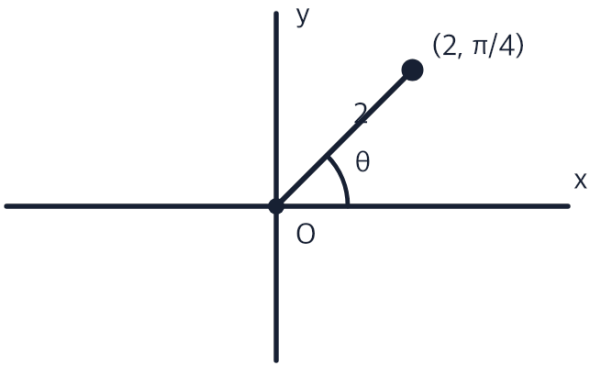
Teacher answers & working

Polar Coordinates and Polar Curves · Polar Coordinates

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SC1

Give equivalent polar coordinates for each point.



Give two different polar coordinates that represent the same point as (2, π/4).

Answer:

(2, 9π/4) and (-2, 5π/4)

Working:

Adding 2π to the angle gives the same ray.

So (2, π/4) is equivalent to (2, 9π/4).

Changing r to -r moves to the opposite ray, so add π to the angle.

So (2, π/4) is also equivalent to (-2, 5π/4).

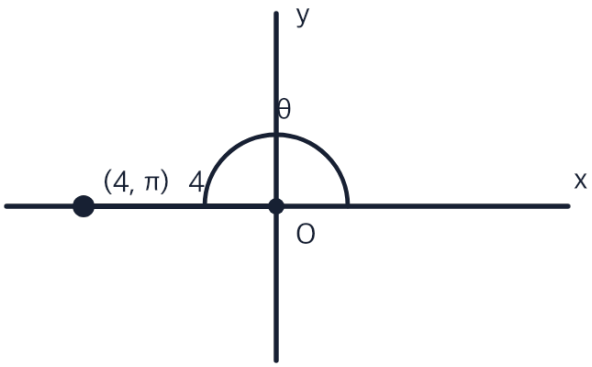
Teacher answers & working

Polar Coordinates and Polar Curves · Polar Coordinates

ALG-294F-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SD1

Convert each polar coordinate into Cartesian form.



Convert the polar coordinate (4, π) into Cartesian form.

Answer:

(-4, 0)

Working:

Use $x = r\cos\theta$ and $y = r\sin\theta$.

Here r = 4 and θ = π.

$x = 4\cos(\pi) = 4 \times \frac{-1}{1} = -4.$

$y = 4\sin(\pi) = 4 \times 0 = 0.$

Therefore the Cartesian coordinate is (-4, 0).

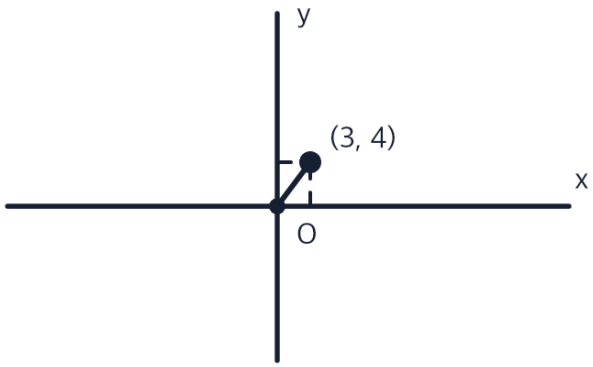
Teacher answers & working

Polar Coordinates and Polar Curves · Polar Coordinates

ALG-294F-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1

Convert each Cartesian coordinate into polar form.



Convert (3, 4) into polar form, using $r > 0$ and $0 \leq \theta < 2\pi$.

Answer:

$(5, \tan^{-1}(4/3))$

Working:

Use $r^2 = x^2 + y^2$.

$r^2 = 3^2 + 4^2 = 25$, so $r = 5$.

Use $\tan\theta = \frac{y}{x} = \frac{4}{3}$.

Choose the quadrant of the point (3, 4).

Therefore $\theta = \tan^{-1}(4/3)$.

The polar coordinate is $(5, \tan^{-1}(4/3))$.

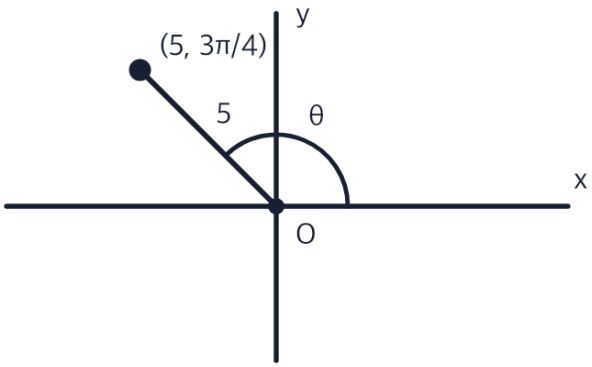
Teacher answers & working

Polar Coordinates and Polar Curves · Polar Coordinates

ALG-294F-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EB1

Give equivalent polar coordinates for each point.



Give two different polar coordinates that represent the same point as (5, 3π/4).

Answer:

(5, 11π/4) and (-5, 7π/4)

Working:

Adding 2π to the angle gives the same ray.

So (5, 3π/4) is equivalent to (5, 11π/4).

Changing r to -r moves to the opposite ray, so add π to the angle.

So (5, 3π/4) is also equivalent to (-5, 7π/4).

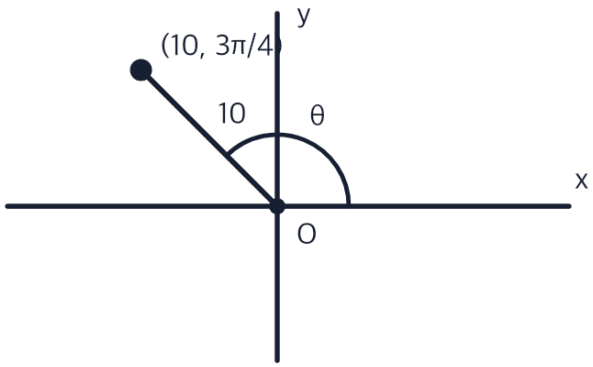
Teacher answers & working

Polar Coordinates and Polar Curves · Polar Coordinates

ALG-294F-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EC1

Convert each polar coordinate into Cartesian form.



Convert the polar coordinate (10, 3π/4) into Cartesian form.

Answer:

$(-5\sqrt{2}, 5\sqrt{2})$

Working:

Use $x = r\cos\theta$ and $y = r\sin\theta$.

Here r = 10 and θ = 3π/4.

$x = 10\cos(3\pi/4) = 10 \times \frac{-\sqrt{2}}{2} = -5\sqrt{2}$.

$y = 10\sin(3\pi/4) = 10 \times \frac{\sqrt{2}}{2} = 5\sqrt{2}$.

Therefore the Cartesian coordinate is $(-5\sqrt{2}, 5\sqrt{2})$.